

10 • Quality Assurance — Risk-Based Testing

Assignment: "Quality Assurance Testing: Design an approach to risk-based testing. Conduct tests to identify L2 – L4 translation issues. Address any bugs or issues identified during testing."

Lecture Day 5: "We cannot test everything, so we test what matters most."

1. Identify Hazards → 2. Rank by Risk → 3. Test Critical Paths First → 4. **Document Residual Risk**

10.1 Testing Approach

We follow the four-step risk-based procedure from Day 5. The depth of testing is determined by the **potential for clinical harm**, not by the number of rules.

Types of Testing Used

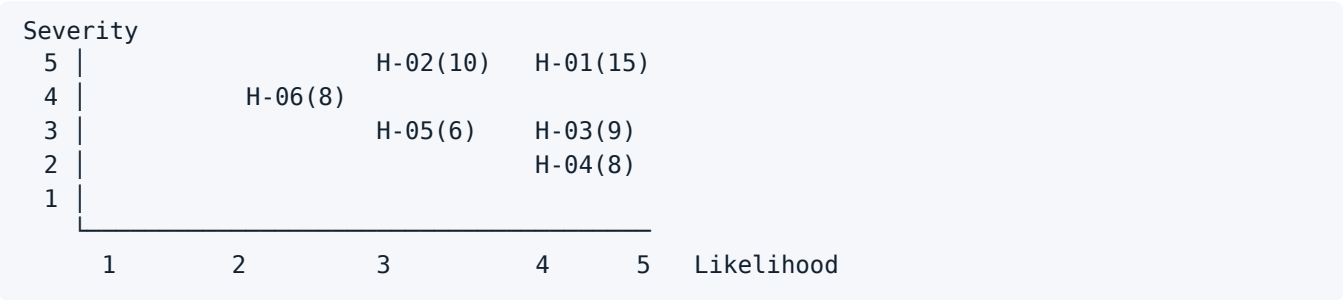
Type (Day 5)	In our project
L2 Verification	Reconciliation of decision table ↔ BPMN gateway cascade ↔ L3 rules: do the conditions, the ordering and the outputs match?
Functional Testing	Individual rules, exclusions, severity ordering, boundary values
Integration Testing	Interplay: derived elements → rules → output; conditional logic → rule reachability; trend rules across several days
Usability Testing	Image assignment by non-clinical persons — not yet carried out (see residual risk)

10.2 Steps 1 & 2 — Hazard Analysis and Prioritisation

ID	Hazard	Origin	Severity (1-5)	Likelih. (1-5)	Score	Mitigation	Covered by
H-01	An infection is classified as GREEN	The rule does not fire; the question is not asked because of conditional logic	5	3	15	Worst-first cascade; safety-critical questions ungated; catch-all only as the final rule	T1-T5, T11-T13
H-02	An emergency is classified as ORANGE instead of RED	Prioritisation incorrect; rule order transposed	5	2	10	Fixed worst-first ordering, documented and tested	T4, T15
H-03	The patient abandons the check, leaving a gap in the trend	Too many questions; excessive duration	3	3	9	Baseline data collected only once; conditional logic reduces the number of questions; ≤ 3 min	T6
H-04	False alarm during normal healing	Threshold set too low; time window not taken into account	2	4	8	Time-dependent rule R6a; information rules actively reassure	T5, T9, T14

ID	Hazard	Origin	Severity (1-5)	Likelih. (1-5)	Score	Mitigation	Covered by
H-05	An implausible input is processed	Constraint missing or applied too late	3	2	6	Constraints run before rule evaluation (validation loop in the BPMN)	T7
H-06	The practice overlooks an alert	Alert fatigue caused by too many notifications	4	2	8	Alerts only for ORANGE/RED; structured content instead of free text	(organisational, not testable)

Risk Matrix



Step 3 — Prioritisation

- **H-01 and H-02** (score ≥ 10): **full coverage**. Every rule at the red and orange level has at least one test case, and all boundary values are checked from both sides.
- **H-03 to H-06** (score 6–9): **sample-based coverage** with 1–2 representative test cases each.
- The green catch-all rule is only tested indirectly — through cases that are *not* supposed to escalate.

10.3 Defects Found

All findings originate from **desk-based testing** and a systematic review of the decision table against the L3. **Three of the five defects arose in the L2 and only became apparent during L4 testing — exactly the “L2-L4 translation issues” that the assignment asks about.**

B-01 • Marked signs of inflammation on days 1-2 are classified as GREEN

Severity	● high
Originating layer	L2 — decision table
Description	R6 was gated with <code>days_post_op >= 3</code> . A patient on day 2 with a markedly reddened, swollen, overheated wound received GREEN via the catch-all rule R8 — provided there was neither fever nor pus
Root Cause	The time-based exception for <i>mild</i> post-operative erythema was inadvertently applied to all signs of inflammation. Related to challenge C-03
Aggravating factor	Test case T5 originally recorded this behaviour as <i>expected</i> — the test would never have found the defect
Fix	R6 split: R6a (mild signs, Southampton I) remains time-dependent · R6b (marked signs, Southampton II or <code>swelling = 3</code>) fires from day 1 . T5 corrected

Retest	T13, T14
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B-02 • "I feel really unwell" triggers nothing

Severity	● high
Originating layer	L2 — decision table
Description	<code>wellbeing</code> was collected (<code>0=Gut</code> · <code>1=Müde</code> · <code>2=Richtig krank</code> — <code>0=well</code> · <code>1=tired</code> · <code>2=really unwell</code>) but appeared in not a single rule . A patient with <code>wellbeing = 2</code> , 37.6 °C and an unremarkable wound received GREEN
Root Cause	The data element was defined but never carried over into the logic. Clinically relevant: a severe subjective sense of illness can precede measurable fever — in early sepsis this is precisely the dangerous pattern
Fix	New rule R9 : <code>wellbeing = 2 OR nausea_vomiting = 2</code> → ORANGE
Retest	T11

B-03 • Emergency rule R2 was unreachable because of conditional logic

Severity	● high
Originating layer	L2/L3 — conditional logic
Description	The question about the extent of erythema was gated with <code>relevant: wound_state >= 2</code> . R2 (spreading erythema / red streak → RED, suspected lymphangitis) could only fire if the patient rated the wound appearance as at least "slightly reddened" ("leicht gerötet"). If they selected "Calm & dry" ("Ruhig & trocken") — plausible when the wound itself is unremarkable but a red streak runs towards the trunk — the question was never asked and the emergency rule was never reached
Root Cause	The conditional logic was set from a UX perspective (fewer questions) without checking the reachability of safety-critical rules. A textbook case of an L2→L4 translation issue
Fix	New, ungated data element <code>red_streak</code> (DE-032): "Does a red streak run away from the wound?" ("Zieht ein roter Streifen von der Wunde weg?"). R2 changed to <code>red_streak = 1 OR redness_extent = 4</code>
Retest	T12
Rule derived from this	Safety-critical questions must never sit behind conditional logic. Adopted as a convention in the L3

B-04 • Duplicated source of truth for the fever threshold

Severity	● medium (maintainability)
Originating layer	L3
Description	The derived element <code>fever = temp >= 38</code> existed but was used nowhere; R5 checked <code>temp >= 38</code> again directly. If a guideline change moved the value to 37.8, two places would have had to be changed — with a high probability of overlooking one of them
Root Cause	A contradiction of our own maintainability claim from 08
Fix	All thresholds as named constants in <code>meta.thresholds</code> ; rules reference <code>fever</code> instead of <code>temp</code>
Retest	T2, T19

B-05 • `anticoag` was collected but never used

Severity	● low (design/data protection)
Originating layer	L2
Description	The anticoagulant status fed into no rule at all — even though it is clinically relevant for assessing bleeding
Root Cause	Collection without a purpose of use. In addition a data protection problem: personal health data collected without a purpose contradict data minimisation
Decision	Build it in rather than remove it — the clinical value justifies collecting it
Fix	New rule R10 : <code>bleeding >= 1 AND anticoag = 1</code> → YELLOW
Retest	T16

B-06 • Southampton grade IV mapped to RED in the L1 table but to ORANGE in the rules

Severity	● medium (inconsistency)
Originating layer	L2 — inconsistency between the L1 mapping table and the decision table
Found by	Test run T15
Description	The Southampton mapping in 03 assigned grade IV (pus) to RED. The implemented rules, however, escalate to RED only for <code>wound_state = 4</code> (the "open / pus" image) or dehiscence (R3); isolated purulent discharge (<code>secretion = 3</code>) triggers R4 → ORANGE. Two documents made different statements about the same finding
Root Cause	The mapping table was derived from the Southampton severity scale ("grades IV and V are major complications"), the rules from the clinical urgency of the action required. Both are defensible — but they were never reconciled
Decision	The rules are authoritative; the mapping table has been corrected. Purulent discharge is the primary criterion of a superficial incisional SSI and requires contact today (ORANGE) — not the emergency service tonight. RED remains reserved for systemic signs (R1), lymphangitis (R2) and dehiscence or the "open / pus" wound image (R3)
Fix	Mapping table in 03 §3.4 corrected; rationale documented as challenge C-09
Retest	T10, T15

B-07 • Trend rule R12 could never fire — circular dependency

Severity	● high (rule without effect)
Originating layer	L3/L4 — evaluation order
Found by	Test run T17
Description	The derived element <code>trend_worsening</code> compares today's classification with yesterday's. But today's classification is only known after the rules have been evaluated — while the derived elements are computed before . <code>classification</code> was therefore always <code>null</code> , <code>trend_worsening</code> always <code>0</code> , and rule R12 could never fire
Root Cause	A classic circular dependency: a rule needs its own result as an input. Only visible once the tests were actually executed — a review of the decision table alone would never have found it
Fix	Two-pass evaluation in the L4: the rules run once, the provisional classification is fed back into the scope, the rules run a second time. The worse of the two results wins, so a second pass can escalate but never de-escalate — worst-first also applies across passes
Retest	T17

B-08 • R12 was unreachable as originally specified

Severity	🟡 medium (dead rule)
Originating layer	L2 — decision table
Found by	Test run T17, after B-07 had been fixed
Description	Even with a working two-pass evaluation, R12 ("deterioration compared with yesterday, yesterday YELLOW") remained unreachable: any deterioration beyond YELLOW already triggers a higher-priority orange or red rule. The condition could only be satisfied in a state that never occurs
Root Cause	The rule was formulated from the clinical intent ("things are getting worse") without checking against the worst-first cascade whether that state is reachable at all
Fix	R12 rewritten as a persistence rule : <code>classification = 'yellow' AND last_classification = 'yellow'</code> → ORANGE. A single abnormal day is tolerable, a persisting abnormal finding is not
Retest	T17
Lesson learned	Every new rule must be checked for reachability against the cascade — added to the L3 review checklist

10.4 Test Protocol

Executed on 3 August 2026 against `l3/wundcheck-l3.json` v0.3.3 using the harness `tools/run-tests.mjs`, which uses the same expression engine as the L4 app. Reproduce with `node tools/run-tests.mjs`.

ID	Checks	Hazard	Expected	Observed	Status
T1	R1 fever	H-01	RED (R1)	RED (R1) · SG 0	✓ passed
T2	R1 boundary	H-01	RED (>= inclusive)	RED (R1) · SG 0	✓ passed
T3	R3 safety rule	H-01	RED (image alone escalates)	RED (R3) · SG V	✓ passed
T4	R5 before R6	H-02	ORANGE (R5 wins)	ORANGE (R5) · SG II	✓ passed
T5	mild signs, day 2	H-04	GREEN	GREEN (R8) · SG I	✓ passed
T6	conditional logic	H-03	3 follow-ups hidden	hidden: true	✓ passed
T7	constraint temp	H-05	input rejected	rejected: true	✓ passed
T8	info rules non-exclusive	—	GREEN + I1 + I2 + I3	GREEN (R8) + I1, I2, I3 · SG 0	✓ passed
T9	scab reassures	H-04	GREEN + I4	GREEN (R8) + I4 · SG 0	✓ passed
T10	worst-first vs reassurance	H-02	ORANGE (R4) + I4 — warning beats reassurance	ORANGE (R4) + I4 · SG IV	✓ passed
T11	B-02 fixed	H-01	ORANGE (R9)	ORANGE (R9) · SG 0	✓ passed

ID	Checks	Hazard	Expected	Observed	Status
T12	B-03 fixed	H-01	RED (R2)	RED (R2) · SG 0	✓ passed
T13	B-01 fixed	H-01	YELLOW (R6b), not green	YELLOW (R6b) · SG II	✓ passed
T14	B-01 counter-check	H-04	GREEN (R6a does not fire)	GREEN (R8) · SG I	✓ passed
T15	Southampton derivation	H-02	grade IV -> ORANGE (R4) — see B-06	ORANGE (R4) · SG IV	✓ passed
T16	B-05 fixed	—	YELLOW (R10)	YELLOW (R10) · SG 0	✓ passed
T17	persistence rule R12	H-01	ORANGE (R12) after a yellow yesterday	ORANGE (R12)	✓ passed
T18	discharge duration R11	H-01	ORANGE (R11)	ORANGE (R11) · SG III	✓ passed
T19	threshold maintainability	—	ORANGE after threshold change	GREEN -> ORANGE (R5)	✓ passed

19 passed, 0 failed, 19 total

Boundary values are covered from both sides: temp 38.9/39.0/39.1 · days_post_op 2/3/4 · pain 6/7 · discharge_duration_days 2/3.

The first run failed 3 of 19 test cases. All three were genuine defects, not harness errors — they became bugs B-06, B-07 and B-08. Two of them (B-07, B-08) were invisible to a pure document review and only surfaced once the rules were actually executed — which is precisely the point of the exercise.

Boundary value tests cover both sides: temp 38.9/39.0/39.1 · days_post_op 2/3/4 · pain 6/7 · discharge_duration_days 2/3.

10.5 FHIR Conformance Checks

The decision logic is not the only thing that has to be correct — so does the interface. tools/check-fhir.mjs extracts the bundle builder straight out of the L4 and checks the generated transaction bundle against the R4 shape rules we depend on. Reproduce with node tools/check-fhir.mjs .

#	Check	Result
F01	Bundle.resourceType = Bundle	✓
F02	Bundle.type = transaction — transaction	✓
F03	every entry has request.method + url	✓
F04	QuestionnaireResponse present	✓
F05	QR.status = completed	✓
F06	QR.questionnaire is versioned canonical — `http://wundcheck.example/Questionnaire/wundcheck`	0.3.2`
F07	QR.subject set — Patient/example	✓
F08	QR items all have exactly one answer	✓

#	Check	Result
F09	QR answers are typed (not all valueString) — <code>valueString</code> , <code>valueDate</code> , <code>valueCoding</code> , <code>valueInteger</code> , <code>valueDecimal</code>	✓
F10	QR item count matches answered elements — 33 / 33	✓
F11	Observation present	✓
F12	Observation.status = final	✓
F13	Observation.code has a SNOMED coding	✓
F14	Observation carries the L3 version — 0.3.2	✓
F15	Observation carries the triggering rule	✓
F16	Communication generated for ORANGE	✓
F17	Communication.priority = urgent for ORANGE — urgent	✓
F18	Communication has a recipient — Organization/practice	✓
F19	endpoint comes from the L3, not the code	✓
F20	no Communication when GREEN	✓

20 passed, 0 failed, 20 total

Check F19 is the interesting one: it asserts that the endpoint URL appears **only** in the L3, not in the app source. Should anyone ever hardcode an endpoint into the L4, this test fails — the maintainability claim is thereby machine-enforced rather than merely asserted in prose.

10.6 No-Hardcoding Check

The central claim of this project — *the L4 contains no clinical content* — is easy to assert and easy to break. We therefore made it a test. `tools/check-hardcoding.mjs` scans the application source (with the embedded L3 payload stripped out) for every element id, section id, rule id, threshold name and value, code system URI, the FHIR endpoint, and a list of clinical vocabulary. Any hit fails the check.

Check	Result
Application source scanned	479 lines
Terms tested (element ids, section ids, rule ids, thresholds, code systems, endpoint, clinical vocabulary)	72
Clinical content found in the L4	✓ none

The check found two real leaks when first run, both of which have been fixed:

Leak	Fix
<code>Observation.code</code> carried SNOMED 225552003 “Assessment of wound” directly in the app, and the result badge read <code>derived.southampton_grade</code> by name	Both declared in the L3 (<code>meta.fhir.observation_code</code> , <code>meta.display.result_badge</code>); the app now iterates over the declaration
The “New day” function reset the sections <code>general</code> and <code>wound</code> by their ids — the app knew which sections were clinical	Reset now derives from the sections’ <code>ONCE</code> flag; the app no longer knows any section by name
Terminology system URIs (<code>http://snomed.info/sct</code> ...) were a lookup table in the app	Moved to <code>meta.terminology_systems</code>

This is the same pattern as B-06: a claim made in prose that the implementation did not actually keep. The difference is that this one can no longer drift silently — the check runs with the test suite.

10.7 Step 4 — Residual Risk

Day 5 explicitly names as a characteristic of sufficient validation: *“Explicit acknowledgment of residual risk and monitoring strategy.”*

What we have not tested:

- (a) **Behaviour with real patients.** All tests are desk-based. We have not had a single real wound assessed.
- (b) **The reliability of image assignment by medical laypersons.** This is the **greatest untested risk of our entire approach**. The translation of a clinical grading into an image selection (C-01) is the central design idea — and it is unvalidated. In particular: erythema appears visually very different on different skin types; our graphics depict only one skin type.
- (c) **Offline and synchronisation behaviour across several days.** The trend rule R12 and the derived elements `last_classification` / `trend_worsening` presuppose an uninterrupted data history. What happens when there are gaps has not been tested.
- (d) **Transferability to all body regions.** The scope covers all elective wounds with primary closure — yet our graphics show only one generic wound type. An abdominal wound looks different from a wound on a finger.
- (e) **Clinical sign-off.** All eight L1→L2 decisions were taken by a group of students, not by clinical professionals.

How We Address This

Residual risk	Measure
(a), (b)	Usability study during the pilot phase; parallel assessment by the patient and a healthcare professional to determine agreement
(b), (d)	Most important safety indicator in post-deployment monitoring: follow-up of all cases that presented to a physician within 72 hours despite a GREEN classification
(c)	Explicit tests of the trend logic with data gaps before the pilot
(d)	Extension of the graphics with body-region-specific variants; until then a scope notice in the app
(e)	Clinical review of all <code>meta.thresholds</code> and decision rules as a precondition for the pilot

Details: 11 Monitoring & Evaluation

10.8 “Good enough validation” — Self-Assessment

Day 5 names six characteristics. Where we stand:

Characteristic	Status
Documented risk analysis identifying highest-harm scenarios	✓ Hazard table H-01...H-06

Characteristic	Status
Evidence of testing on critical clinical paths	✓ 19 of 19 test cases executed and passed; 8 defects found and fixed
Clinical expert sign-off on rule logic and edge cases	✗ missing — documented as a limitation
Usability evidence that users can act correctly	✗ missing — greatest residual risk
Post-deployment monitoring plan with incident response	✓ 11 M&E
Explicit acknowledgment of residual risk	✓ Section 10.7

Four out of six. The two missing points are exactly those that separate a student prototype from a deployable system — and both require access to clinical staff and to real patients.

Related documents: 06 Decision Logic · 07 Challenges · 11 M&E